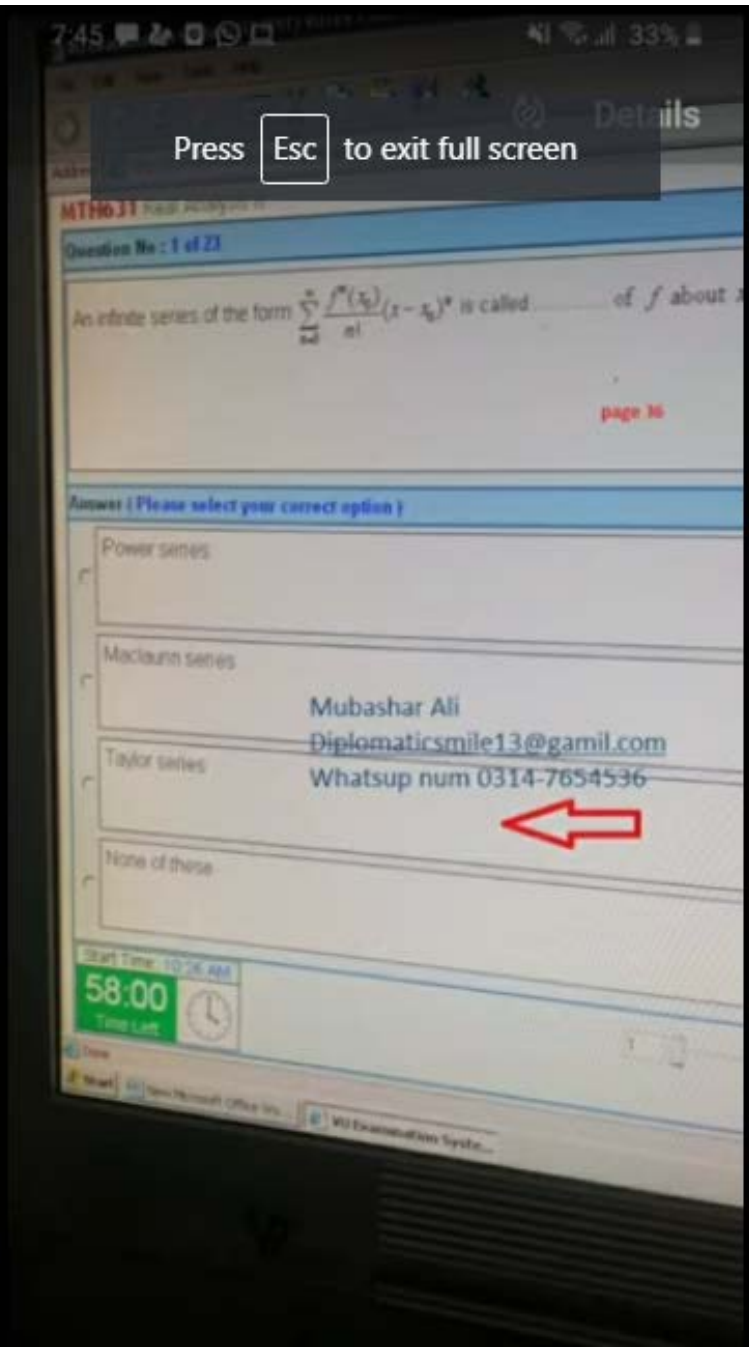
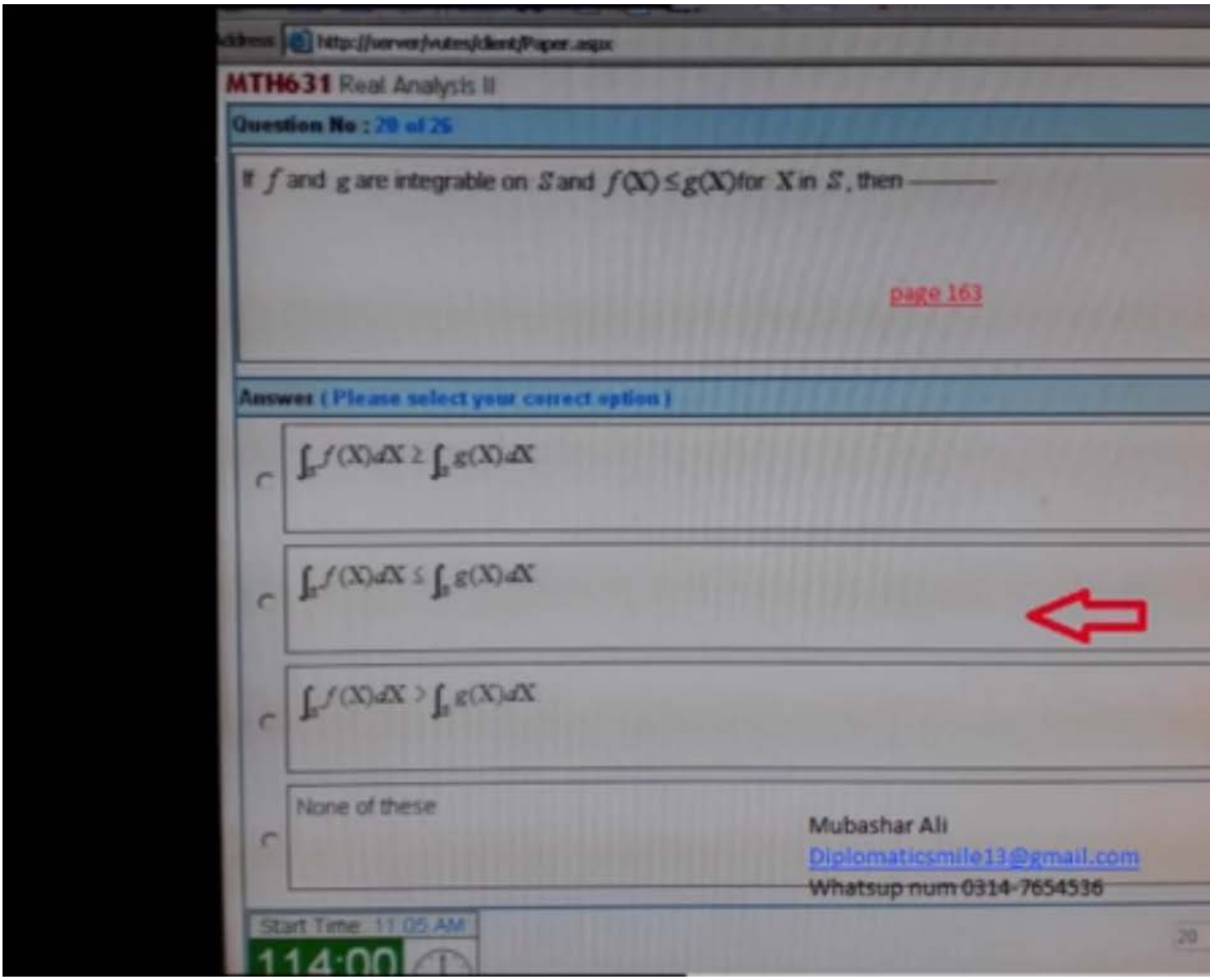
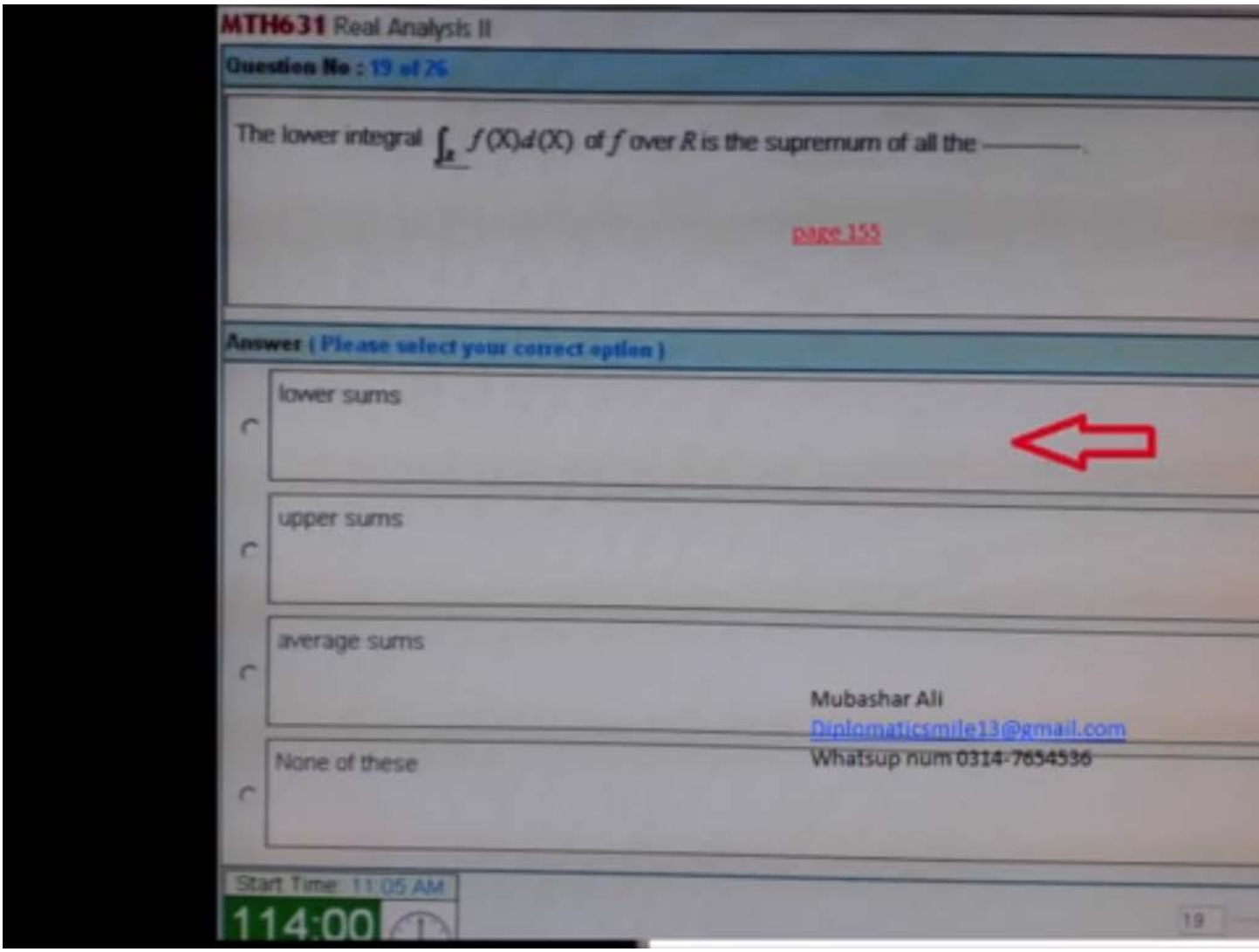
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Question No : 1 of 23
An infinite series of the form $\sum \frac{f^n(x_0)}{n!}(x-x_0)^n$ is called ... of $f$ about
page 36
Answer ( Please select your correct option )
Power series
Maclaurin series
Taylor series
None of these
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58:00


MTH631 Real Analysis II
Question No : 20 of 25
If $f$ and $g$ are integrable on $S$ and $f(X) \leq g(X)$ for $X$ in $S$, then ———
page 163
Answer ( Please select your correct option )
$\int_S f(X)dX \geq \int_S g(X)dX$
$\int_S f(X)dX \leq \int_S g(X)dX$
$\int_S f(X)dX > \int_S g(X)dX$
None of these
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MTH631 Real Analysis II
Question No : 19 of 26
The lower integral $\underline{\int_R} f(X)d(X)$ of $f$ over $R$ is the supremum of all the ———.
page 155
Answer ( Please select your correct option )
lower sums
upper sums
average sums
None of these
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19

**MTH631** Real Analysis II

**Question No :** 18 of 25

The upper integral $\overline{\int_R} f(X)d(X)$ of $f$ over $R$ is the infimum of all the_____

page 135

**Answer** ( Please select your correct option )

- lower sums
- upper sums
- average sums
- None of these



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**MTH631** Real Analysis II

**Question No : 17 of 26**

A function $f$ is said to be absolutely integrable on $[a,b)$ if $f$ is locally integrable on $[a,b)$ and ———

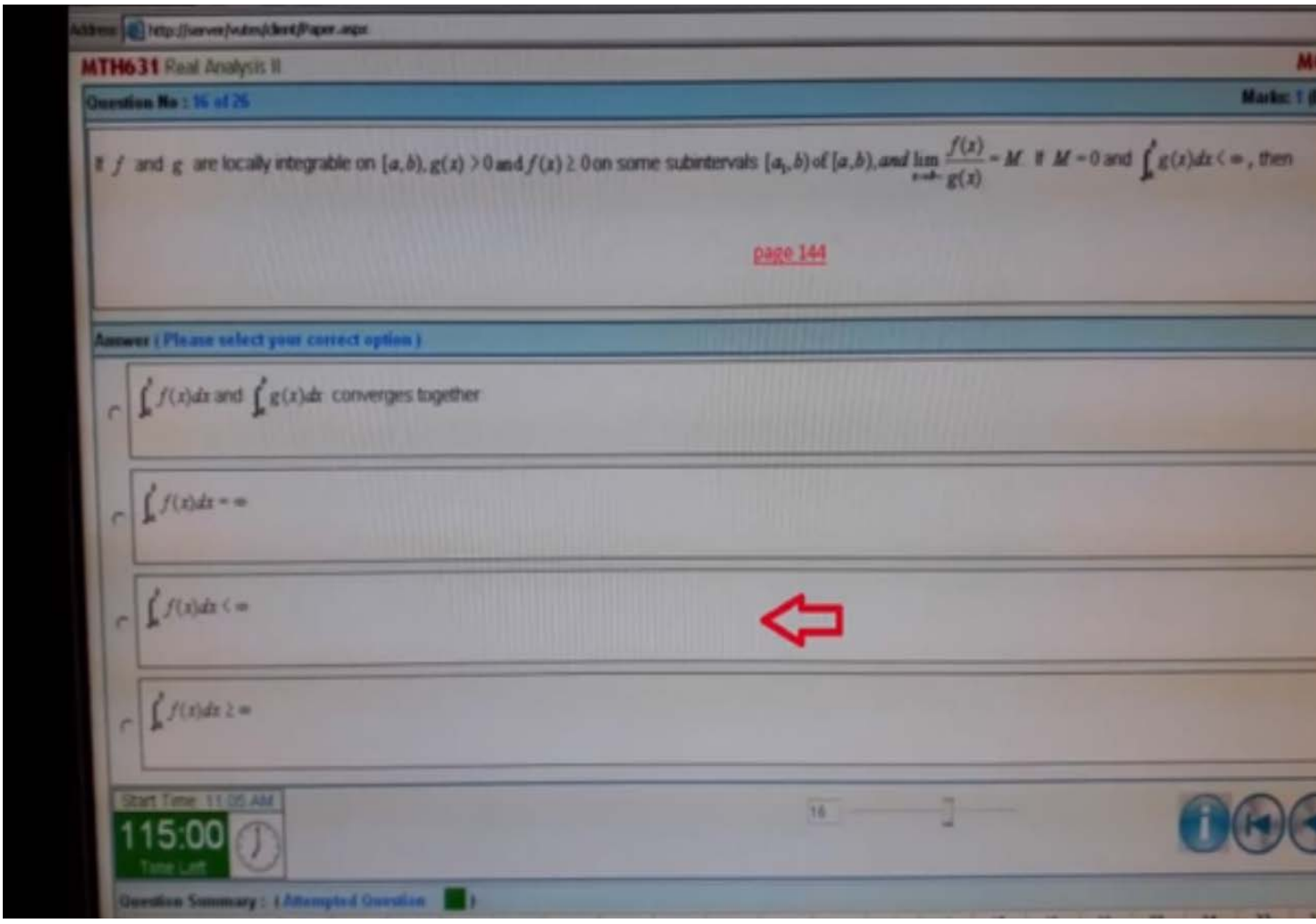
page 145

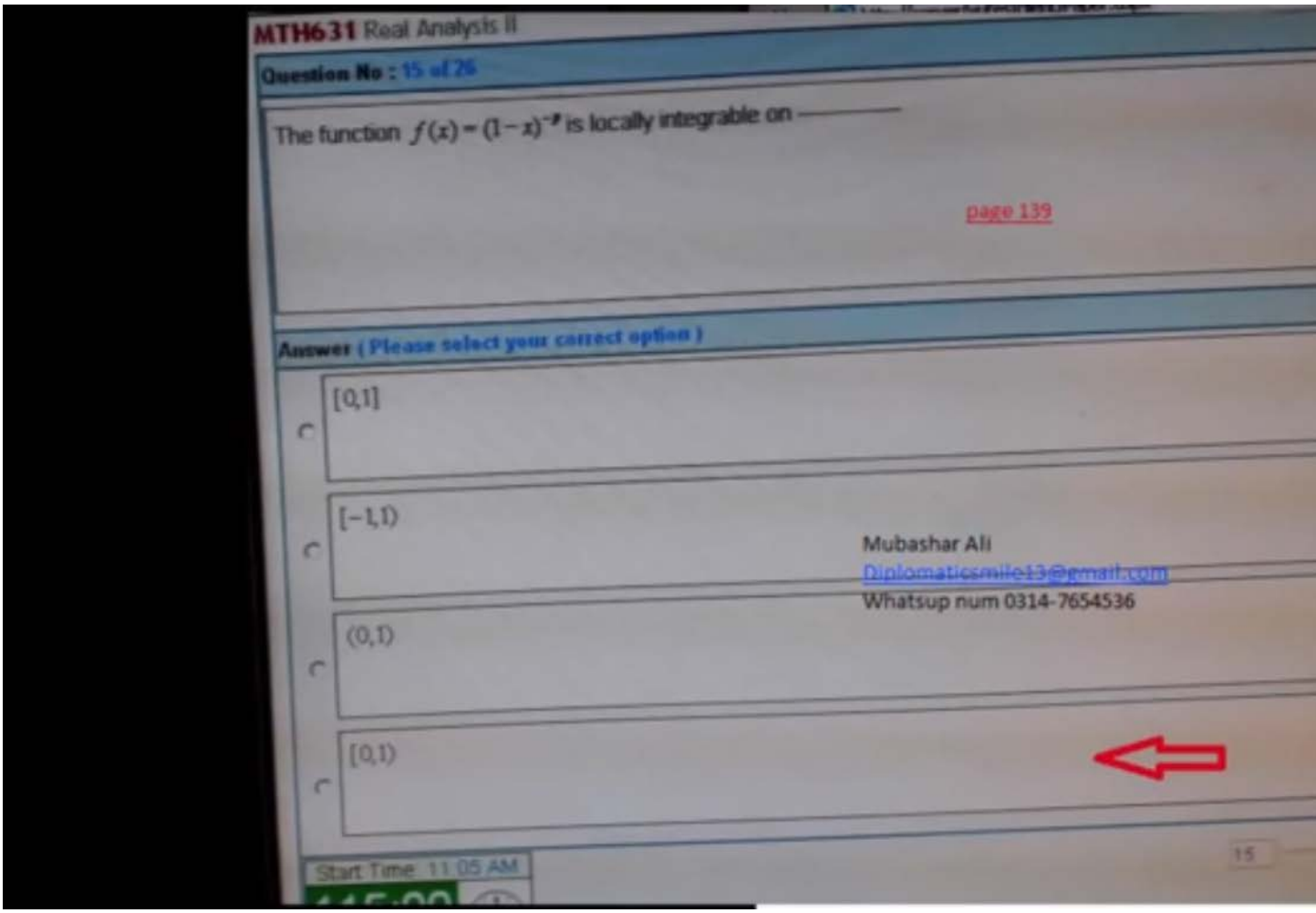
**Answer ( Please select your correct option )**

- $\int_a^b |f(x)|dx < \infty$
- $\int_a^b |f(x)|dx = \infty$
- $\int_a^b |f(x)|dx \geq \infty$
- None of these

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MTH631 Real Analysis II
If $f$ and $g$ are locally integrable on $[a,b)$, $g(x)>0$ and $f(x)\geq 0$ on some subintervals $[a_1,b)$ of $[a,b)$, and $\lim_{x\to b^-}\frac{f(x)}{g(x)}=M$ If $M=0$ and $\int_a^b g(x)dx<\infty$, then
page 144
Answer ( Please select your correct option )
$\int_a^b f(x)dx$ and $\int_a^b g(x)dx$ converges together
$\int_a^b f(x)dx=\infty$
$\int_a^b f(x)dx<\infty$
$\int_a^b f(x)dx\geq\infty$
Start Time: 11:05 AM
115:00
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16


MTH631 Real Analysis II
Question No : 15 of 26
The function $f(x) = (1-x)^{-p}$ is locally integrable on _____
page 139
Answer ( Please select your correct option )
[0,1]
[-1,1)
(0,1)
[0,1)
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15

**MTH631** Real Analysis II

**Question No : 11 of 26**

If $p(X)$ is a homogeneous polynomial of degree $r$ in $X - X_0$ If $p(X) \leq 0$ for all $X \neq X_0$ Then $p$ is called _____

page 134

**Answer ( Please select your correct option )**

- positive semidefinite
- positive definite
- negative semidefinite
- negative definite

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**MTH631** Real Analysis II

**Question No : 13 of 26**

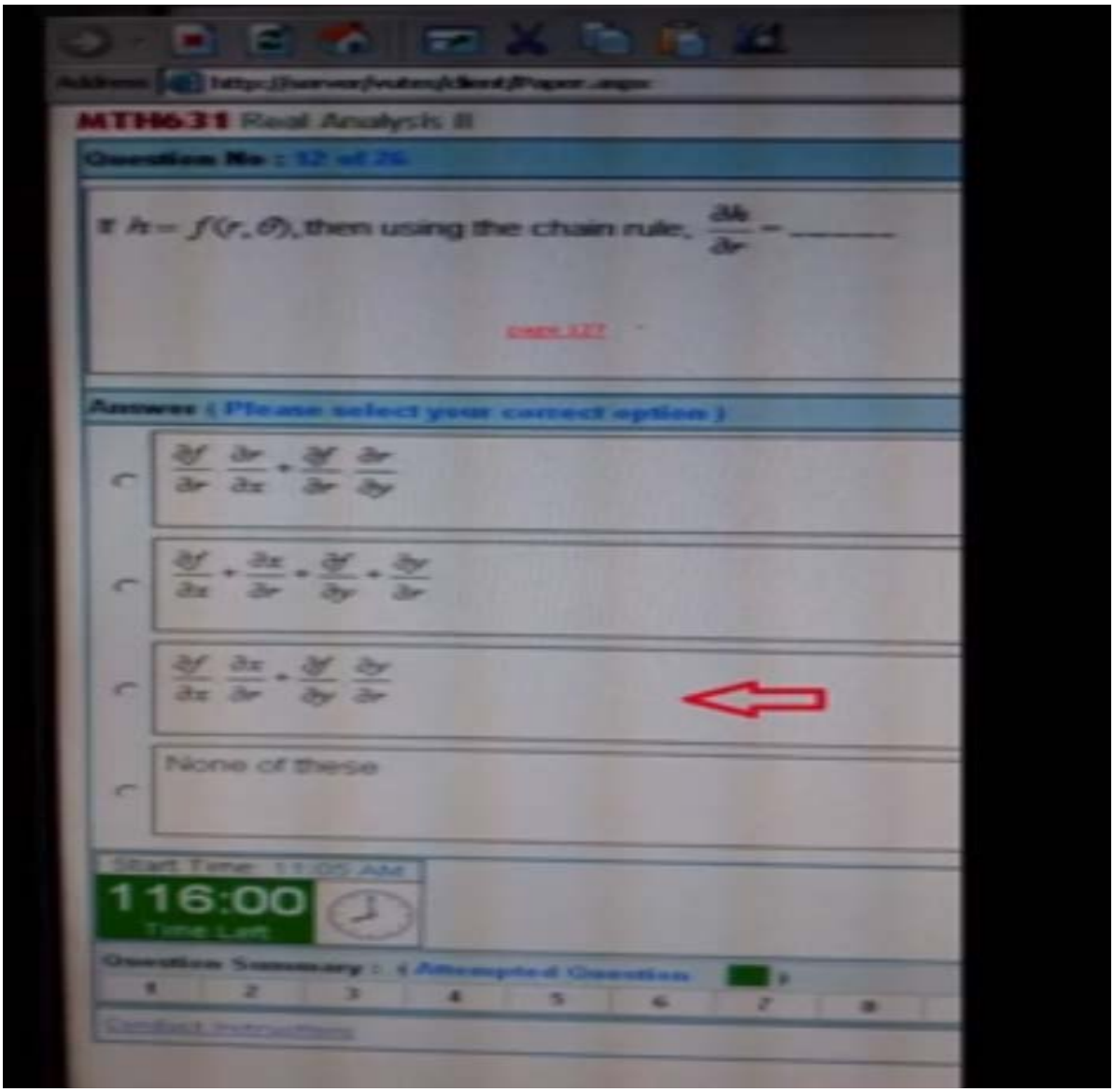
If value of a becomes zero in the Taylor series $\sum_{n=0}^{\infty} \frac{f^n(a)}{n!}(x-a)^n$, then the series is called ———

page 44

**Answer ( Please select your correct option )**

- Maclaurin series
- Fourier series
- Power series
- None of these

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MTH631 Real Analysis II
Question No : 32 of 36
If $h = f(r, \theta)$, then using the chain rule, $\frac{\partial h}{\partial r} =$ _____
Answer ( Please select your correct option )
$\frac{\partial f}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial r}$
$\frac{\partial f}{\partial x} + \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} + \frac{\partial y}{\partial r}$
$\frac{\partial f}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial r}$
None of these
116:00
Time Left
Question Summary :
Attempted Question
1
2
3
4
5
6
7
8

**MTH631** Real Analysis II

**Question No : 11 of 26**

If $f(x) = 3x + 5x^3$, then at $x = -1$, $df =$ _____

**Answer ( Please select your correct option )**

- $18dy$
- $18dx$
- 18



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**MTH631** Real Analysis II

**Question No : 10 of 26**

A function of the form $L(X) = m_1 x_1 + m_2 x_2 + \ldots + m_n x_n$ for some constants $m_i$'s is called ———

page 112

**Answer ( Please select your correct option )**

- linear function
- quadratic function
- constant function
- None of these

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**MTH631** Real Analysis II

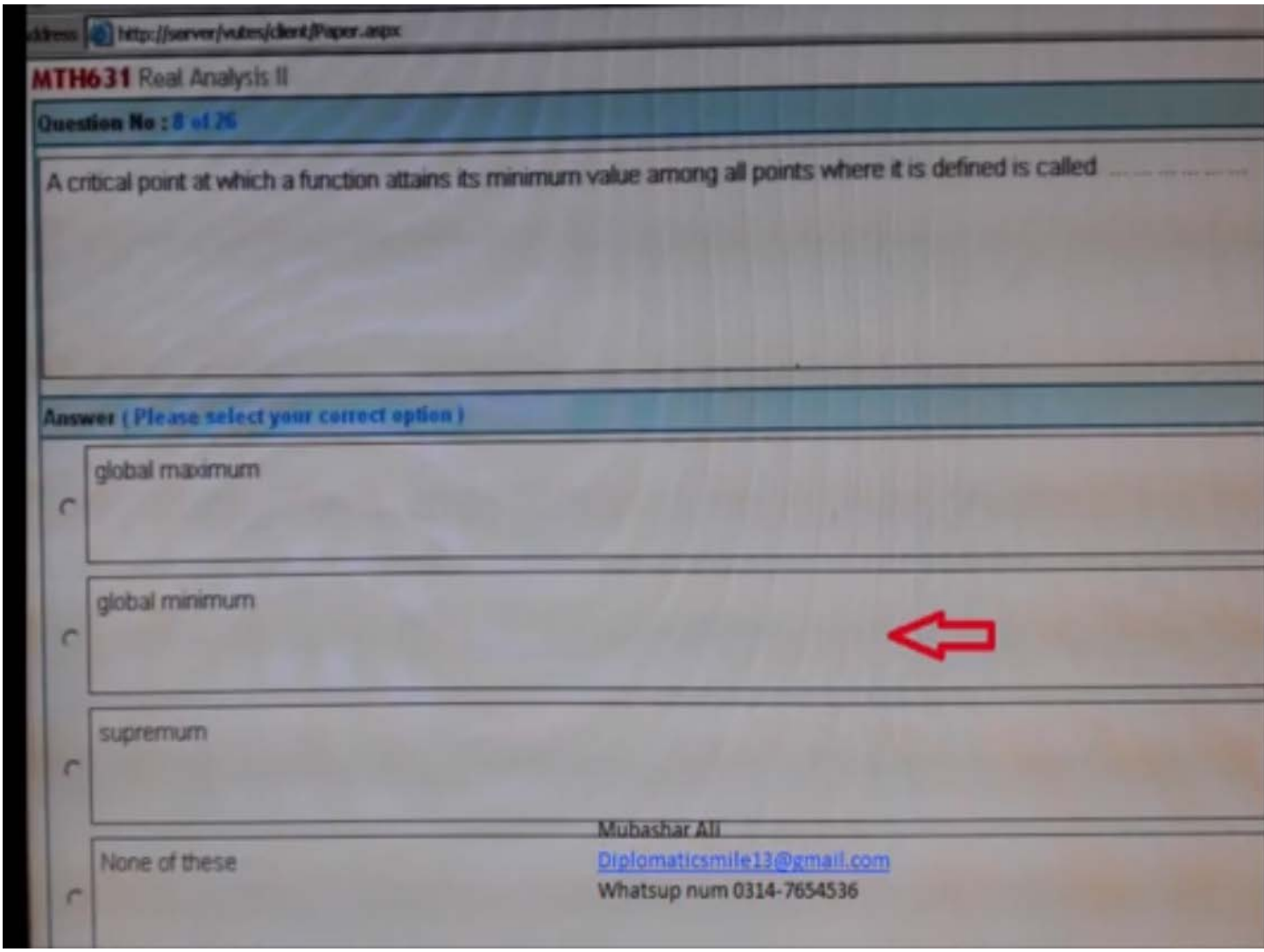
**Question No : 9 of 26**

For the function $f(x,y) = 3x^2y^3 + xy$ in two variables, $f_{xy}(0,0) =$ .....

**Answer ( Please select your correct option )**

- 1
- -1
- 0
- None of these

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MTH631 Real Analysis II
Question No : 8 of 26
A critical point at which a function attains its minimum value among all points where it is defined is called
Answer ( Please select your correct option )
global maximum
global minimum
supremum
None of these
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**MTH631** Real Analysis II

**Question No : 7 of 26**

A $f_1, f_2, \ldots f_k$ exist on a neighborhood of $X_0$ and are continuous at $X_0$, then $f$ is ——— at $X_0$.

page 115

**Answer ( Please select your correct option )**

- uniformly converget
- piecewise convergent
- differentiable
- None of these

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**MTH631** Real Analysis II

**Question No : 6 of 26**

A set A ⊆ R of real numbers is ———————if there exists a real number M ⊆ R, such that $x \leq M$ for every $x \in A$.

**Answer ( Please select your correct option )**

- bounded below
- uniformly continuous
- bounded above
- None of these

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**MTH631** Real Analysis II

**Question No :** 5 of 26

In a vector valued function $G=(g_1,g_2,\ldots,g_n)$. Then $g_1,g_2,\ldots,g_n$ are called the ———— functions of $G$.

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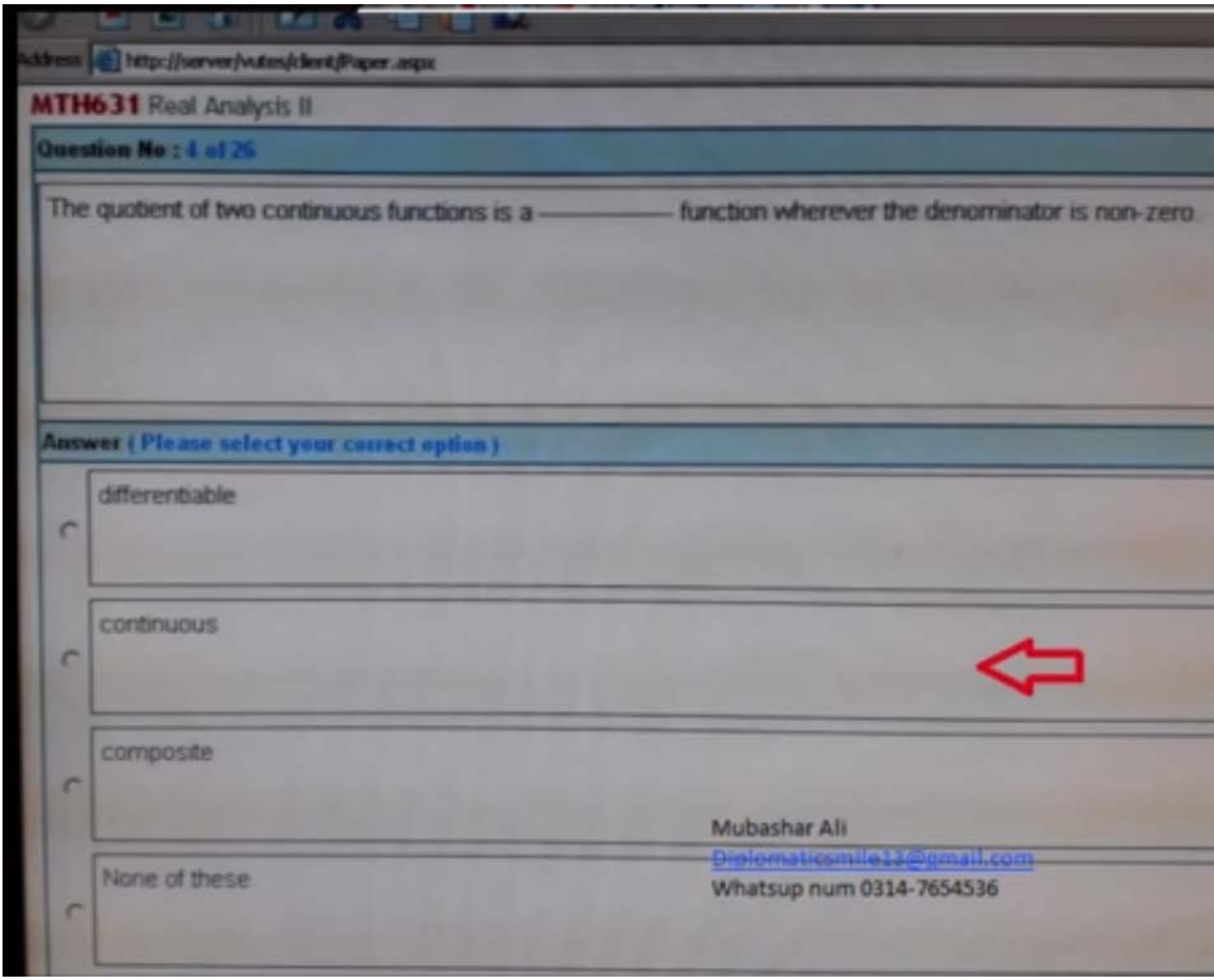
**Answer ( Please select your correct option )**

- bijective
- surjective
- component
- None of these

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MTH631 Real Analysis II
Question No : 4 of 26
The quotient of two continuous functions is a ———— function wherever the denominator is non-zero
Answer ( Please select your correct option )
differentiable
continuous
composite
None of these
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**MTH631** Real Analysis II

**Question No : 2 of 26**

Any polygonally connected set either it is open or not, is always _____

page 82

**Answer ( Please select your correct option )**

- connected 
- disconnected
- a region

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**MTH631** Real Analysis II

**Question No : 3 of 26**

The set $S = \{(x,y), x^2 + y^2 \leq 1 \text{ or } x^2 + y^2 \geq 4\}$ is not a region in $R^2$, since it is _____

page 90

**Answer ( Please select your correct option )**

- not connected
- connected
- polygonally connected
- None of these

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**MTH631** Real Analysis II

**Question No : 2 of 25**

A set S is polygonally connected if every pair of points in S can be connected by a polygonal path lying ————————

page 89

**Answer ( Please select your correct option )**

- outside the set S
- partially in S
- entirely in S
- None of these

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**MTH631** Real Analysis II

**Question No : 1 of 26**

A set is closed if and only if it always contains all of its ——

page 86

**Answer ( Please select your correct option )**

- limit points 
- neighbourhood points
- interior points
- None of these

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MTH631 Real Analysis II

Question No : 26 of 26

Suppose that $f(x,y)=\begin{cases} 0 & \text{if x and y are rational,} \\ 1 & \text{if x is rational and y is irrational,} \\ 2 & \text{if x is irrational and y is rational,} \\ 3 & \text{if x and y are irrational.} \end{cases}$

Find

$$\overline{\int_R} f(x,y)\,d(x,y) \quad \text{and} \quad \underline{\int_R} f(x,y)\,d(x,y) \quad \text{if} \quad R=[a,b]\times[c,d].$$

Let $\mathbf{P}=\{R_1,R_2,\ldots,R_k\}$. Let

$$M_j=\sup\{f(x,y)\,|\,\mathbf{X}\in R_j\} \quad \text{and} \quad m_j=\inf\{f(x,y)\,|\,\mathbf{X}\in R_j\}.$$

Let $j$ be arbitrary in $\{1,2,\ldots,k\}$. Since $R_j$ contains a point $(\hat{x}_j,\hat{y}_j)$ with $\hat{x}_j$ and $\hat{y}_j$ irrational, $M_j=3$. Hence, $\overline{\int_R} f(x,y)\,d(x,y)=3(b-a)(d-c)$. Since $R_j$ contains a point $(\tilde{x}_j,\tilde{y}_j)$ with $\tilde{x}_j$ and $\tilde{y}_j$ rational, $m_j=0$. Hence, $\underline{\int_R} f(x,y)\,d(x,y)=0$.

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**MTH631** Real Analysis II

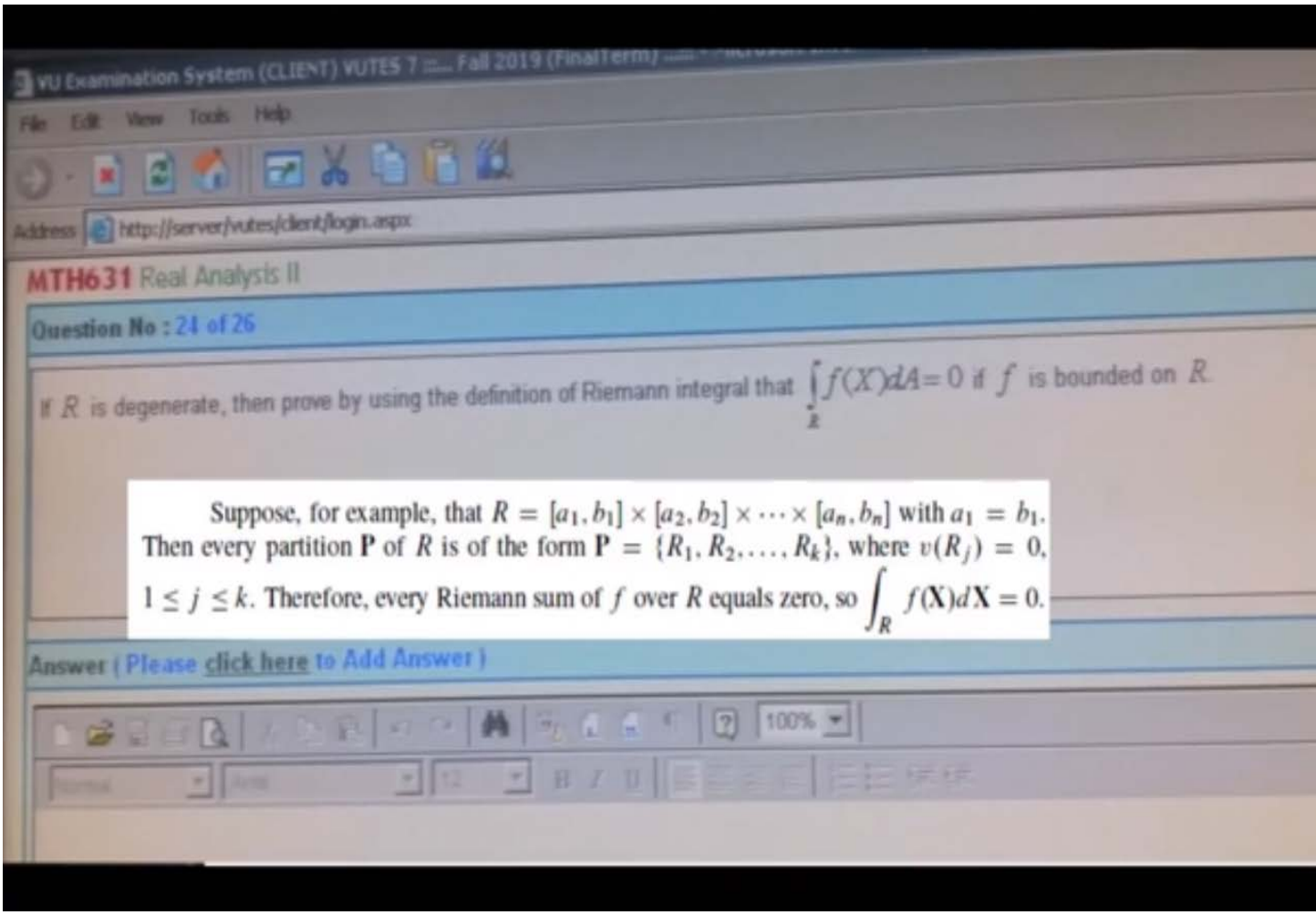
**Question No : 25 of 26**

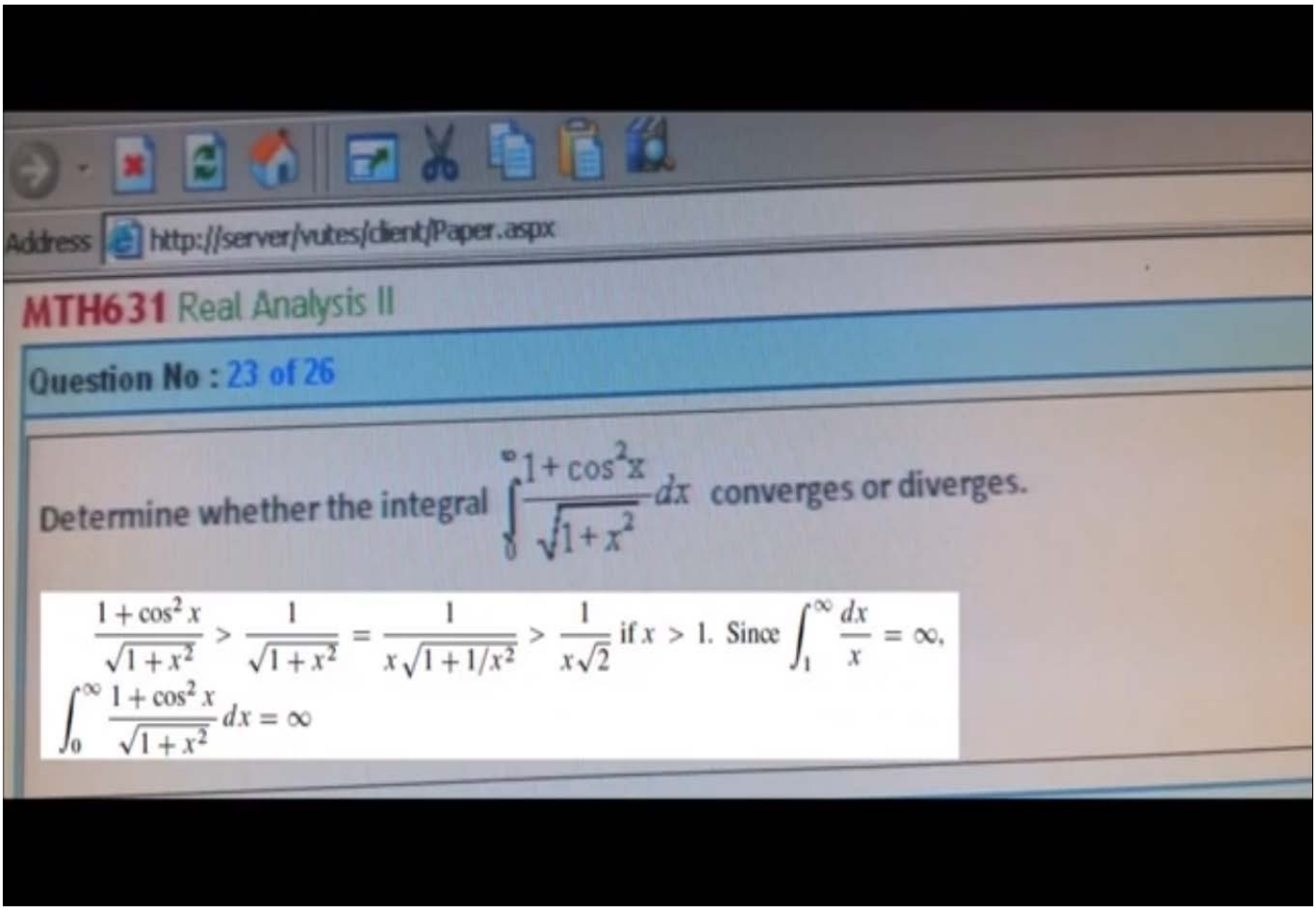
Prove: If a nonempty subset of $R^n$ is both open and closed, then $S=R^n$.

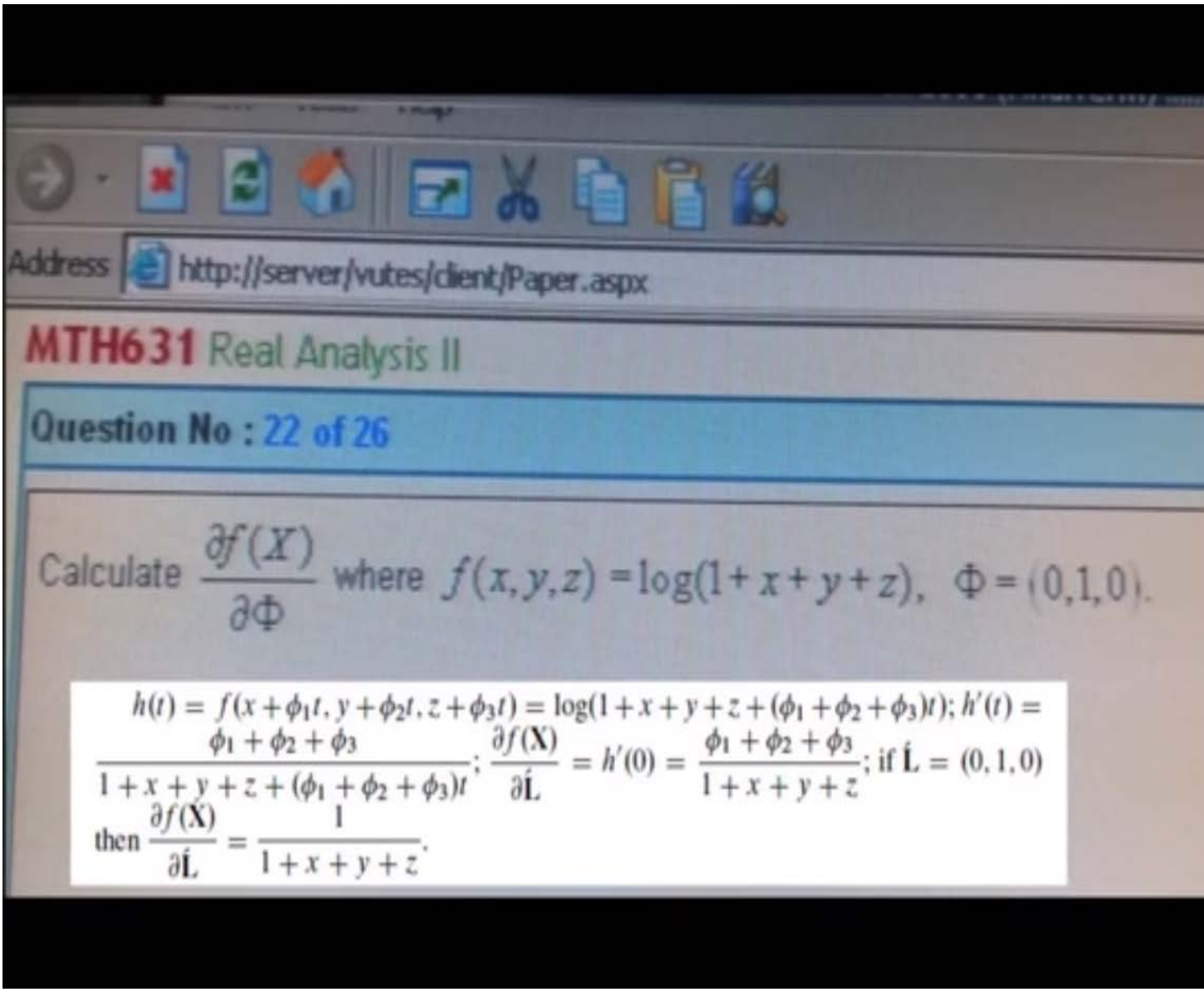
Suppose that $\mathbf{X}_0$ is in $S$. If $S \neq \mathbb{R}^n$, there is an $\mathbf{X}_1 \notin S$. Let

$$H = \{\tau \mid (1-t)\mathbf{X}_0 + t\mathbf{X}_1 \in S \text{ for } 0 \leq t < \tau\}.$$

Since $S$ is open it contains a neighborhood of $\mathbf{X}_0$, so $H \neq \emptyset$. Since $\mathbf{X}_1$ is not in $S$, $\tau \leq 1$ for all $\tau$ in $H$. Let $\rho = \sup H$ and $\overline{\mathbf{X}} = (1-\rho)\mathbf{X}_0 + \rho\mathbf{X}_1$; then $\overline{\mathbf{X}}$ is a limit point of $S$ and so in $S$, because $S$ is closed. Since $S$ is open, it contains some $\epsilon$−neighborhood of $\overline{\mathbf{X}}$, so $\rho + \epsilon/2$ is in $H$. This contradicts the definition of $\rho$. Hence, $S = R^n$.


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MTH631 Real Analysis II
Question No : 24 of 26
If $R$ is degenerate, then prove by using the definition of Riemann integral that $\int_R f(X)dA = 0$ if $f$ is bounded on $R$.
Suppose, for example, that $R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n]$ with $a_1 = b_1$. Then every partition $\mathbf{P}$ of $R$ is of the form $\mathbf{P} = \{R_1, R_2, \ldots, R_k\}$, where $v(R_j) = 0$, $1 \leq j \leq k$. Therefore, every Riemann sum of $f$ over $R$ equals zero, so $\int_R f(\mathbf{X})d\mathbf{X} = 0$.
Answer ( Please click here to Add Answer )
100%


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MTH631 Real Analysis II
Question No : 23 of 26
Determine whether the integral $\int_0^\infty \frac{1+\cos^2 x}{\sqrt{1+x^2}}dx$ converges or diverges.
$\frac{1+\cos^2 x}{\sqrt{1+x^2}} > \frac{1}{\sqrt{1+x^2}} = \frac{1}{x\sqrt{1+1/x^2}} > \frac{1}{x\sqrt{2}}$ if $x > 1$. Since $\int_1^\infty \frac{dx}{x} = \infty$,
$\int_0^\infty \frac{1+\cos^2 x}{\sqrt{1+x^2}}dx = \infty$


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MTH631 Real Analysis II
Question No : 22 of 26
Calculate $\frac{\partial f(X)}{\partial \Phi}$ where $f(x,y,z) = \log(1+x+y+z)$, $\Phi = (0,1,0)$.
$h(t) = f(x+\phi_1 t, y+\phi_2 t, z+\phi_3 t) = \log(1+x+y+z+(\phi_1+\phi_2+\phi_3)t)$; $h'(t) = \frac{\phi_1+\phi_2+\phi_3}{1+x+y+z+(\phi_1+\phi_2+\phi_3)t}$; $\frac{\partial f(\mathbf{X})}{\partial \acute{\mathbf{L}}} = h'(0) = \frac{\phi_1+\phi_2+\phi_3}{1+x+y+z}$; if $\acute{\mathbf{L}} = (0,1,0)$ then $\frac{\partial f(\mathbf{X})}{\partial \acute{\mathbf{L}}} = \frac{1}{1+x+y+z}$.

Question No : 20 of 26

A ———— surface in $\mathbb{R}^n$ has zero content.

PAGE 161

Answer ( Please select your correct option )

- parabolic
- elliptical
- differentiable
- None of these

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MTH631 Real Analysis II

Question No : 19 of 26

If $f$ is integrable on a rectangle $R$, then ———

page 158

Answer ( Please select your correct option )

- $\overline{\int_R} f(x)d(x) > \underline{\int_R} f(x)d(x) = \int_R f(x)d(x)$
- $\overline{\int_R} f(x)d(x) = \underline{\int_R} f(x)d(x) = \int_R f(x)d(x)$ ⇦
- $\overline{\int_R} f(x)d(x) > \underline{\int_R} f(x)d(x) > \int_R f(x)d(x)$
- None of these

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**MTH631** Real Analysis II

**Question No : 18 of 26** Marks: 1

The upper integral $\overline{\int_a^b} f(X)d(X)$ of $f$ over $R$ is the infimum of all the———

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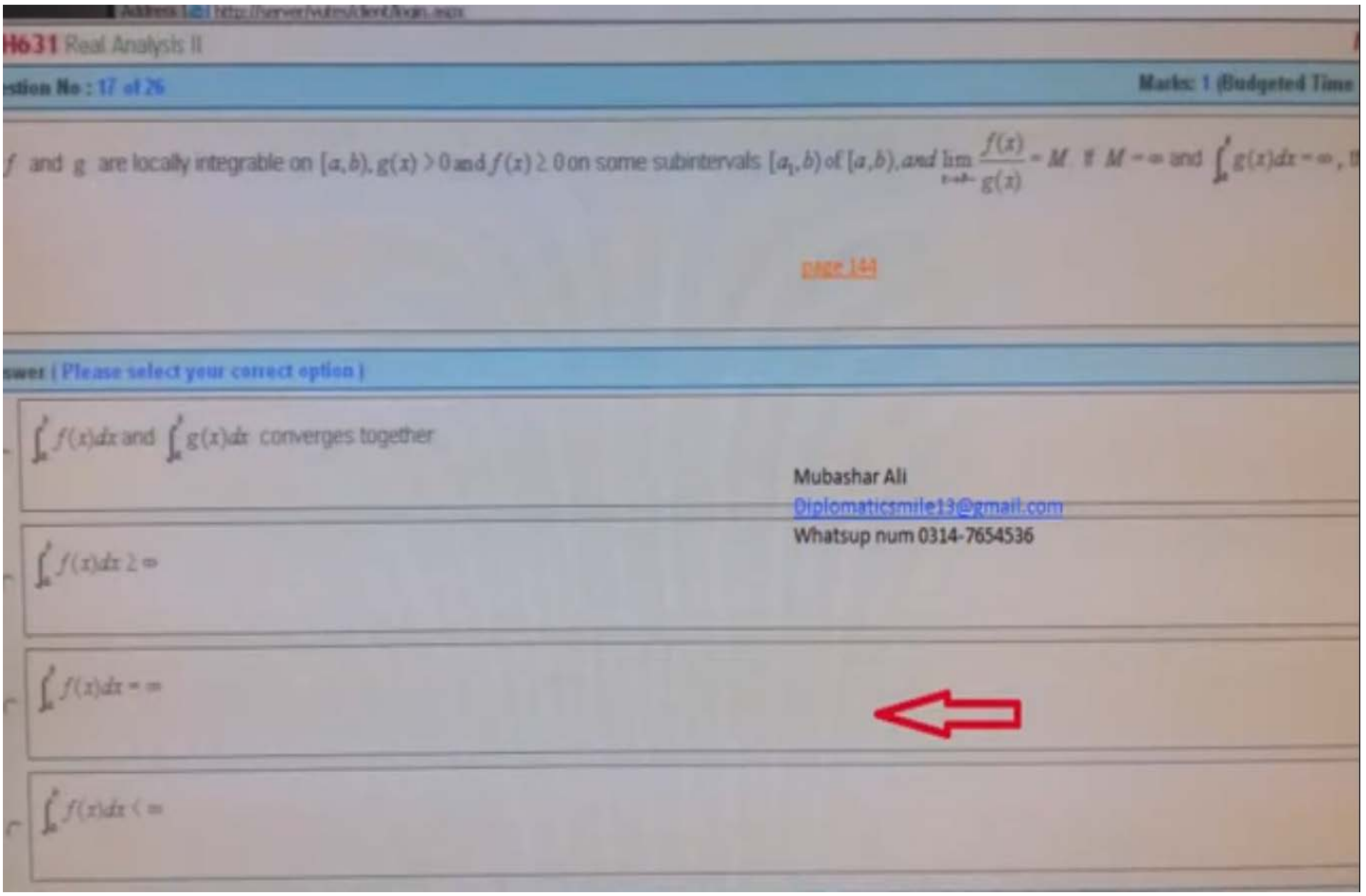
**Answer ( Please select your correct option )**

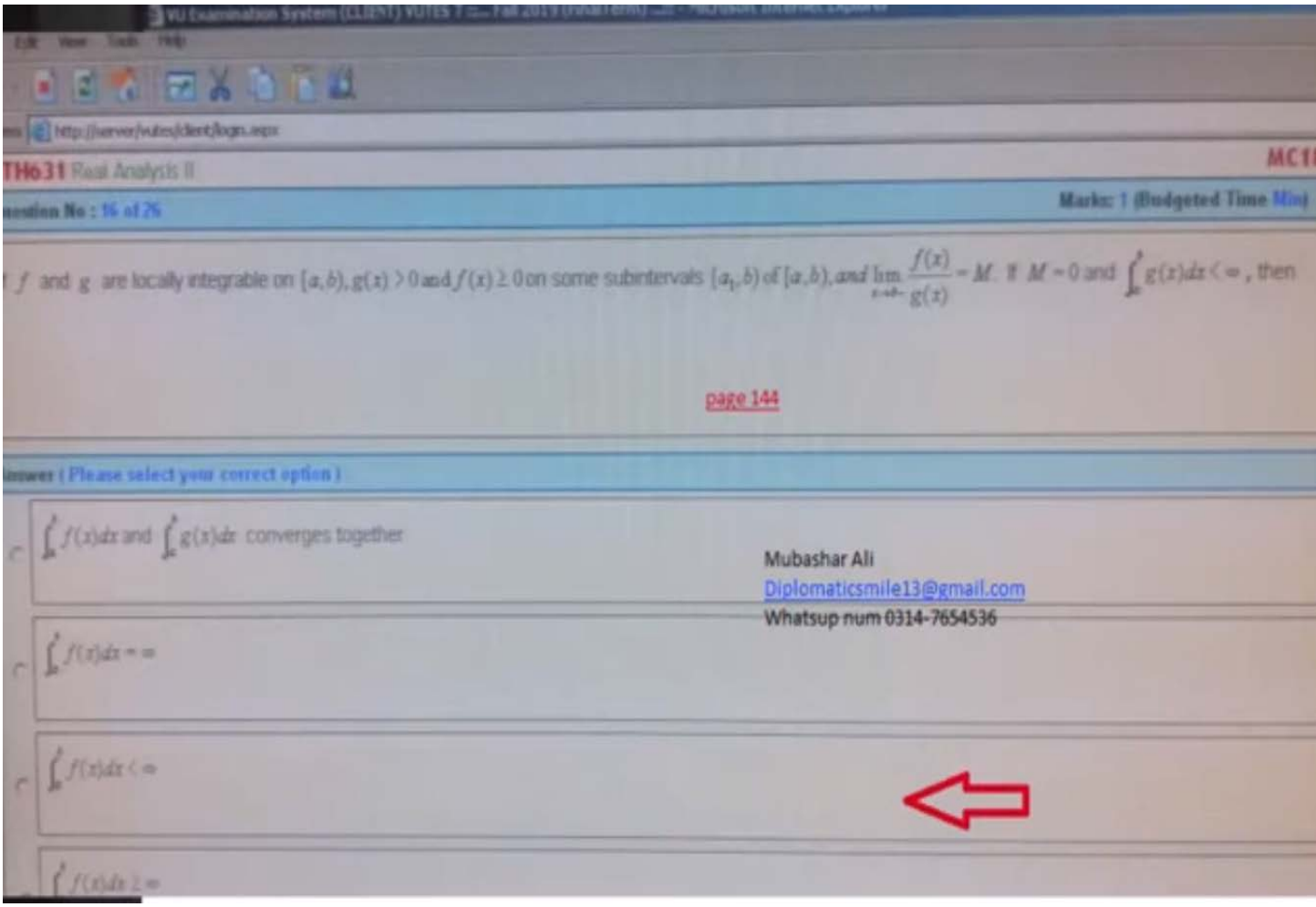
- lower sums
- upper sums
- average sums
- None of these



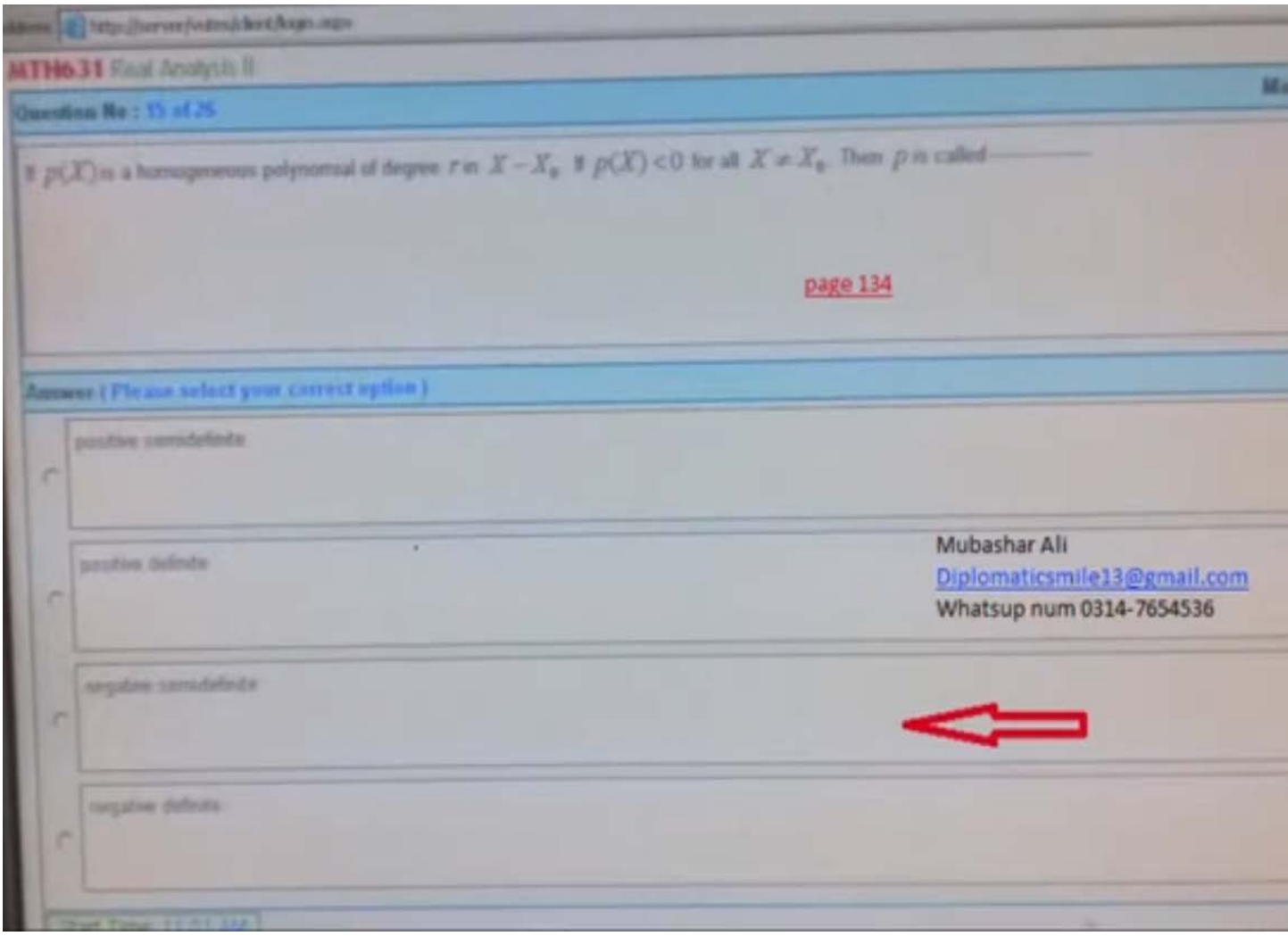
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H631 Real Analysis II
stion No : 17 of 26
Marks: 1 (Budgeted Time
f and g are locally integrable on [a,b), g(x) > 0 and f(x) ≥ 0 on some subintervals [a1,b) of [a,b), and lim f(x)/g(x) = M. If M = ∞ and ∫ g(x)dx = ∞,
page 144
swer ( Please select your correct option )
∫ f(x)dx and ∫ g(x)dx converges together
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∫ f(x)dx ≥ ∞
∫ f(x)dx = ∞
∫ f(x)dx < ∞




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MTH631 Real Analysis II
Question No : 15 of 25
If $p(X)$ is a homogeneous polynomial of degree $r$ in $X - X_0$ if $p(X) < 0$ for all $X \neq X_0$. Then $p$ is called
page 134
Answer ( Please select your correct option )
positive semidefinite
positive definite
negative semidefinite
negative definite
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**Question No : 11 of 26**

If $p(X)$ is a homogeneous polynomial of degree $r$ in $X - X_0$. If $p(X) \geq 0$ for all $X$, then $p$ is called ———

page 134

**Answer ( Please select your correct option )**

- positive semidefinite
- positive definite
- negative semidefinite
- negative definite



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Question No : 13 of 26

If value of a becomes zero in the Taylor series $\sum_{n=0}^{\infty} \frac{f^n(a)}{n!}(x-a)^n$, then the series is called ———

page 44

Answer ( Please select your correct option )

- Maclaurin series
- Fourier series
- Power series
- None of these

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**MTH631** Real Analysis II

**Question No : 12 of 26**

If $h = f(x,y), x = f(r,\theta)$ and $y = f(r,\theta)$, then using the chain rule, $\frac{\partial h}{\partial \theta} =$ _____

page 127

**Answer ( Please select your correct option )**

- $\frac{\partial f}{\partial r}\frac{\partial r}{\partial \theta} + \frac{\partial f}{\partial r}\frac{\partial r}{\partial \theta}$
- $\frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \theta}$
- $\frac{\partial f}{\partial x} + \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} + \frac{\partial y}{\partial \theta}$

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Question No : 11 of 26

If $f(x)=3x+5x^3$, then at $x=-1$, $df=$ _____

Answer ( Please select your correct option )

- $18dy$
- $18dx$
- 18

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MTH631 Real Analysis II

Question No : 10 of 26

A function of the form $L(X) = m_1 x_1 + m_2 x_2 + \ldots + m_n x_n$ for some constants $m_i$'s is called ———

page 112

Answer ( Please select your correct option )

- linear function 
- quadratic function
- constant function
- None of these

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MTH631 Real Analysis II

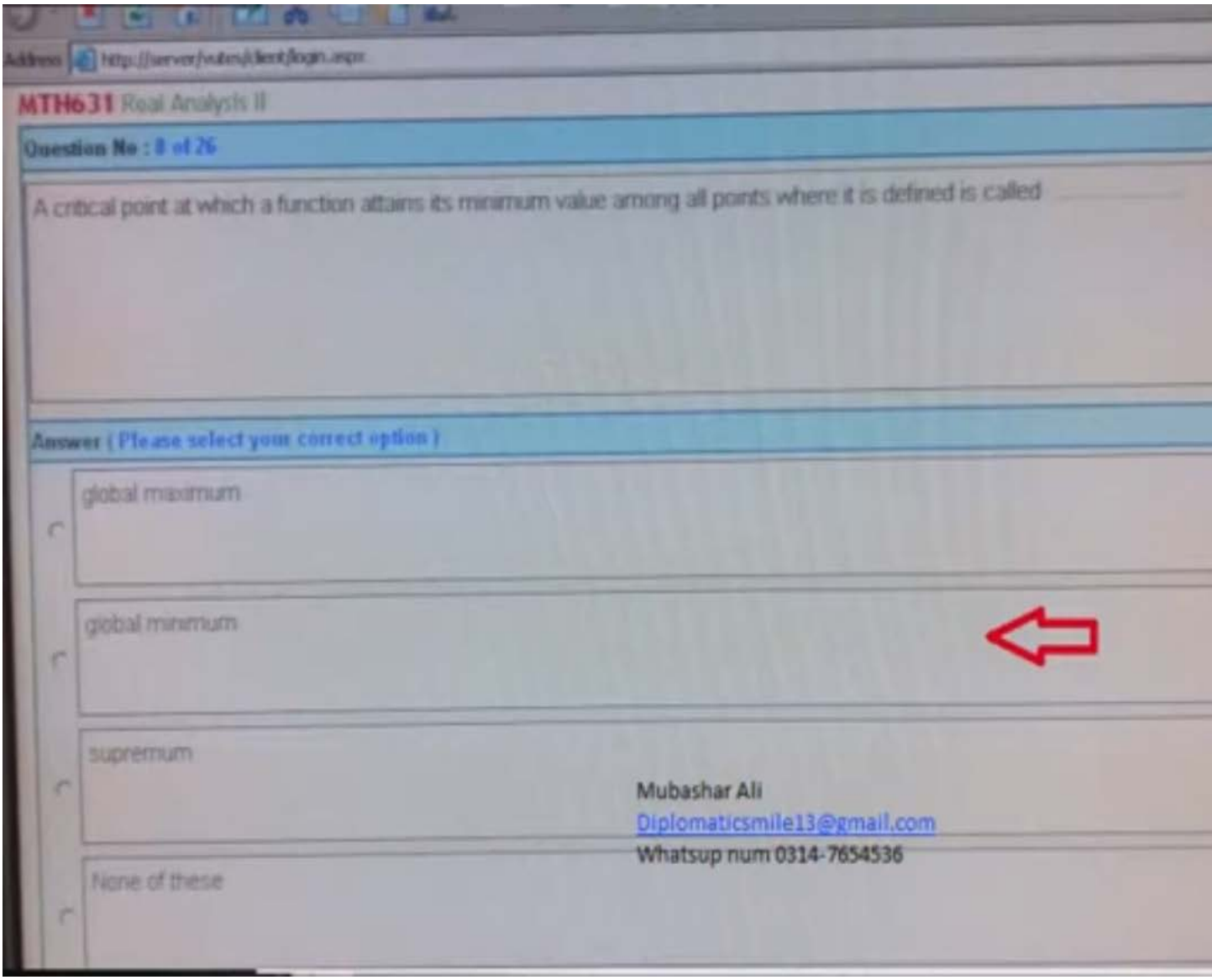
Question No : 9 of 26

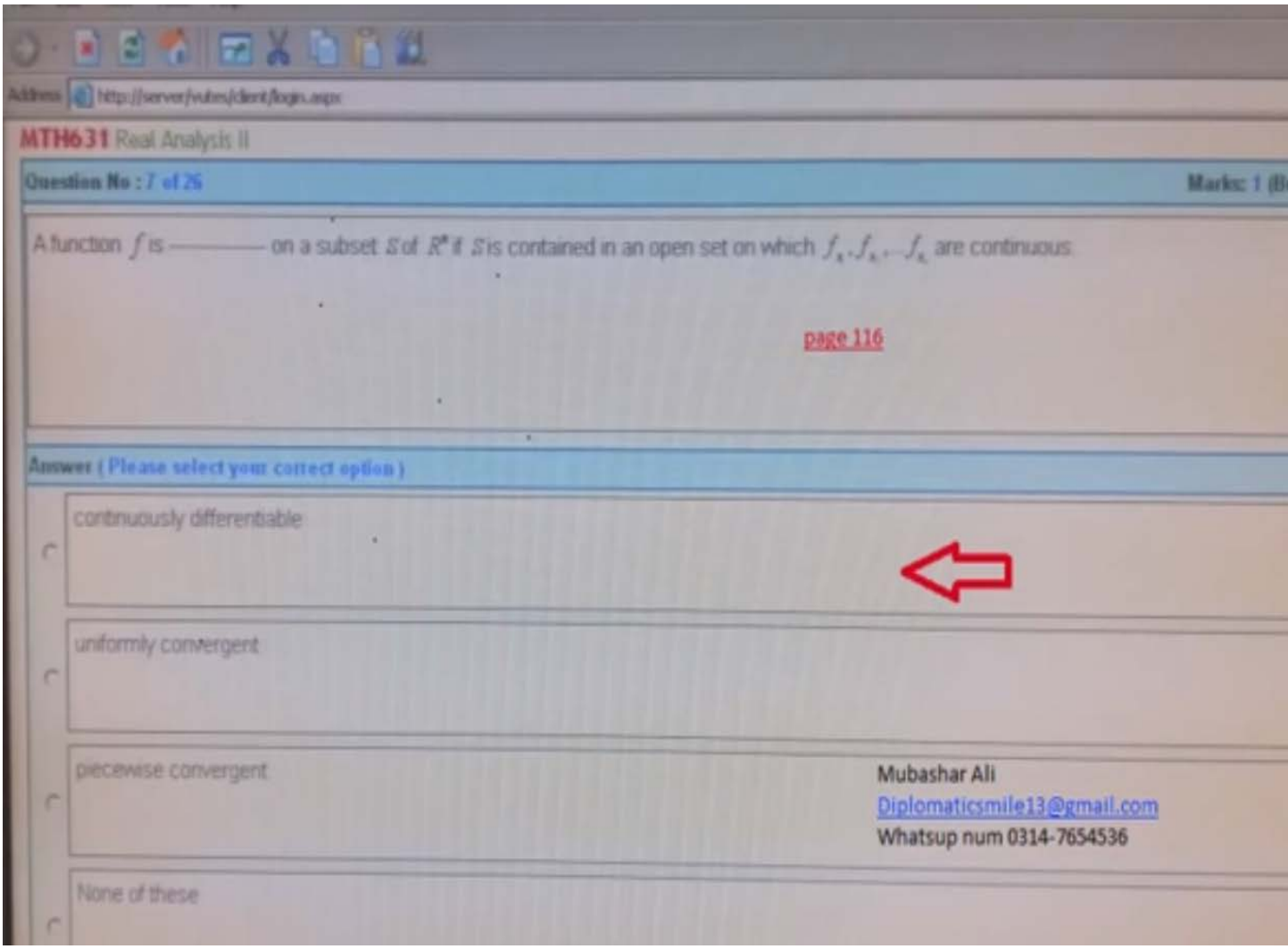
The rate of change of $f(x,y)$ in the direction of a unit vector is called the ———— of $f(x,y)$

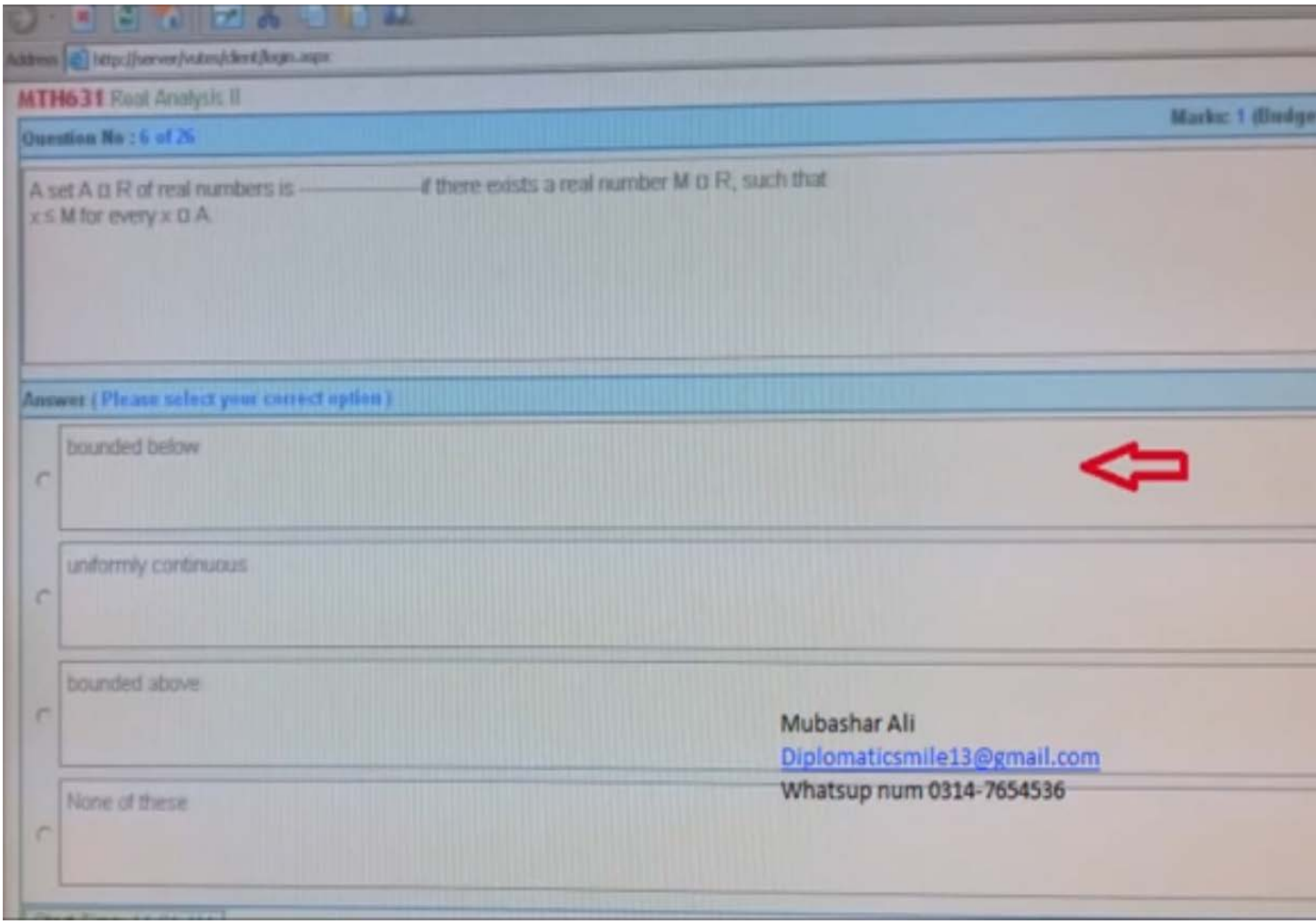
Answer ( Please select your correct option )

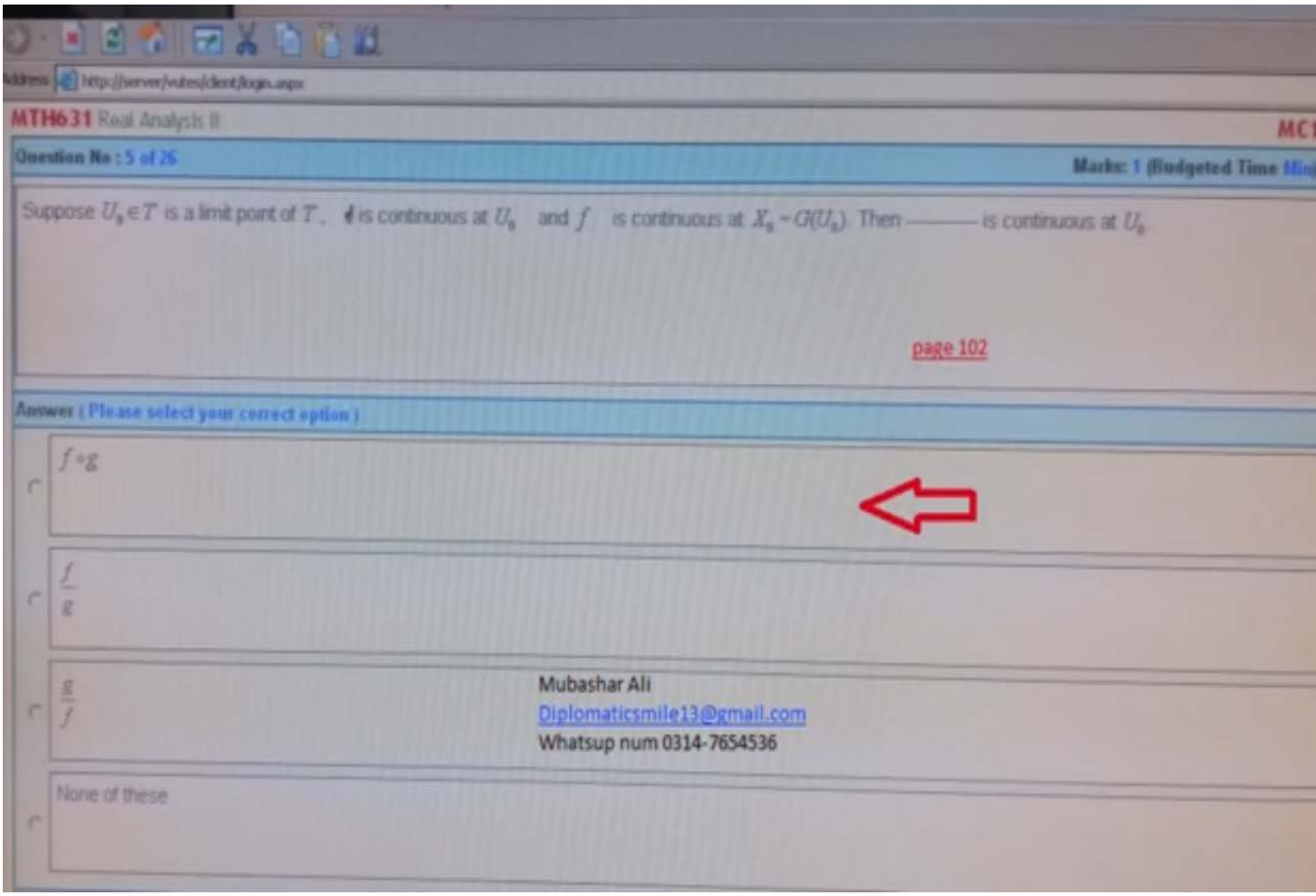
- differential
- directional derivative
- mixed derivative
- None of these

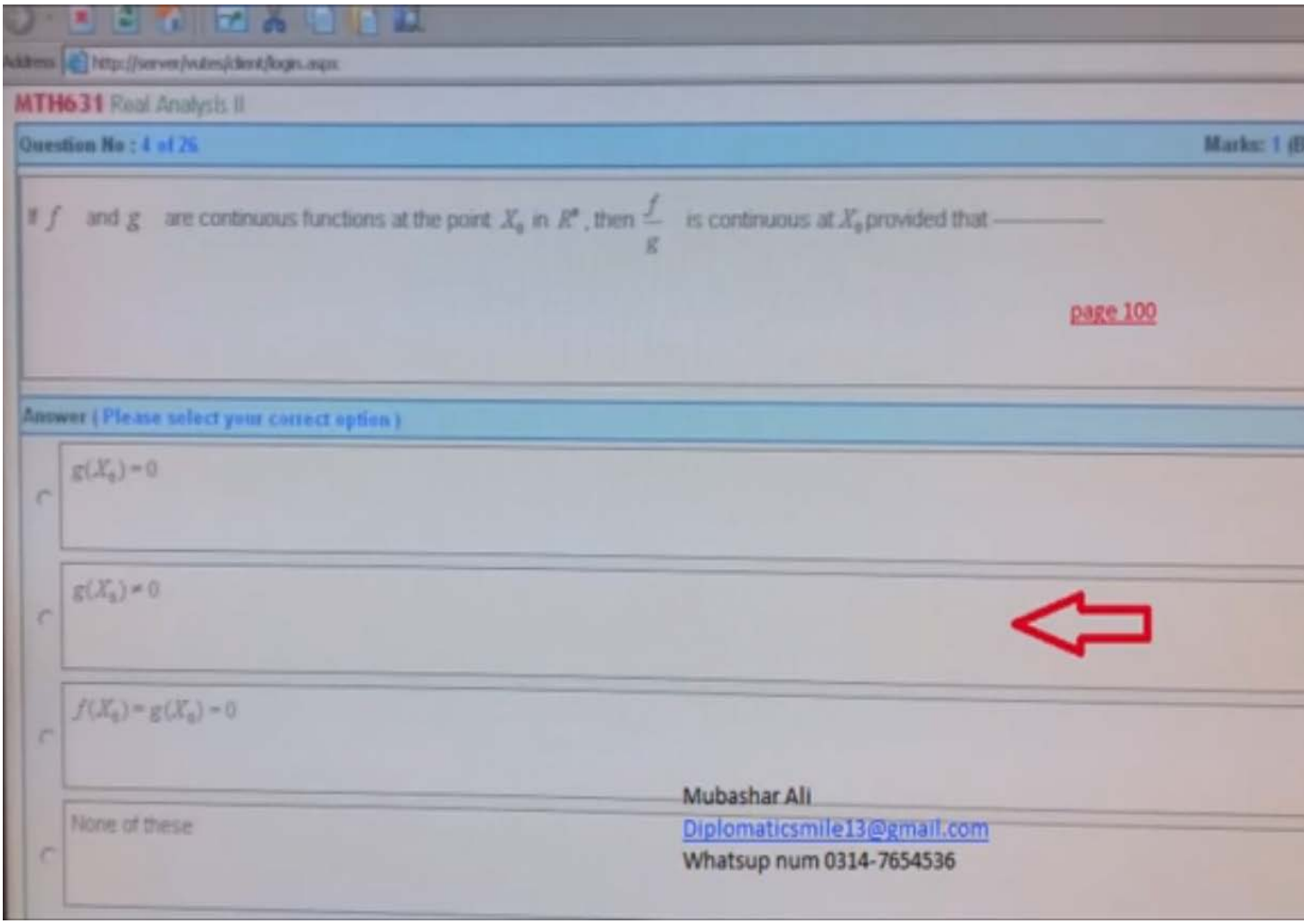
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MTH631 Real Analysis II
Question No : 8 of 26
A critical point at which a function attains its minimum value among all points where it is defined is called
Answer ( Please select your correct option )
global maximum
global minimum
supremum
None of these
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MTH631 Real Analysis II
Question No : 7 of 26
Marks: 1
A function $f$ is ———— on a subset $S$ of $R^n$ if $S$ is contained in an open set on which $f_x, f_y, \ldots, f_n$ are continuous
page 116
Answer ( Please select your correct option )
continuously differentiable
uniformly convergent
piecewise convergent
None of these
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MTH631 Real Analysis II
Question No : 6 of 26
A set A ⊆ R of real numbers is ---------- if there exists a real number M ∈ R, such that x ≤ M for every x ∈ A.
Answer ( Please select your correct option )
bounded below
uniformly continuous
bounded above
None of these
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MTH631 Real Analysis II
Question No : 5 of 26
Marks: 1 (Budgeted Time
Suppose $U_0 \in T$ is a limit point of $T$. $\phi$ is continuous at $U_0$ and $f$ is continuous at $X_0 = G(U_0)$. Then ——— is continuous at $U_0$
page 102
Answer ( Please select your correct option )
$f \circ g$
$\frac{f}{g}$
$\frac{g}{f}$
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None of these


MTH631 Real Analysis II
Question No : 4 of 26
Marks: 1
If $f$ and $g$ are continuous functions at the point $X_0$ in $R^n$, then $\frac{f}{g}$ is continuous at $X_0$ provided that _____
page 100
Answer ( Please select your correct option )
$g(X_0) = 0$
$g(X_0) \neq 0$
$f(X_0) = g(X_0) = 0$
None of these
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MTH631 Real Analysis II

Question No : 3 of 26 Marks

For the function $f(x,y)=\frac{xy}{x^2+y^2}$, the limit of $f(x,y)$ as $(x,y)\to(0,0)$ along the line $y=x$ is ——

page 87

Answer ( Please select your correct option )

- $\frac{1}{2}$
- $-\frac{1}{2}$
- 0
- 1

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**MTH631** Real Analysis II

**Question No : 18 of 25**

A function of the form $L(X) = m_1 x_1 + m_2 x_2 + \ldots + m_n x_n$ for some constants $m_i$'s is called ———

page 112

**Answer ( Please select your correct option )**

- linear function
- quadratic function
- constant function
- None of these

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**MTH631** Real Analysis II

**Question No : 9 of 25**

For the function $f(x,y) = 3x^2y^3 + xy$ in two variables, $f_{xy}(0,0) =$.....



**Answer ( Please select your correct option )**

- 1
- -1
- 0
- None of these

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**MTH631** Real Analysis II

**Question No : 8 of 25**

A critical point at which a function attains its minimum value among all points where it is defined is called ... ... ... ... ... ...

**Answer ( Please select your correct option )**

- global maximum
- global minimum
- supremum
- None of these

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**MTH631** Real Analysis II

**Question No : 7 of 26**

A $f_1, f_2, \ldots f_n$ exist on a neighborhood of $X_0$ and are continuous at $X_0$. then $f$ is ——— at $X_0$

page 115




**Answer ( Please select your correct option )**

- uniformly converget
- piecewise convergent
- differentiable
- None of these

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**MTH631** Real Analysis II

**Question No : 6 of 26**

A set A ⊂ R of real numbers is _____ if there exists a real number M ⊂ R, such that $x \leq M$ for every $x \subset A$.

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**Answer ( Please select your correct option )**

- bounded below
- uniformly continuous
- bounded above
- None of these

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**MTH631** Real Analysis II

**Question No : 5 of 26**

In a vector valued function $G = (g_1, g_2, \ldots, g_n)$. Then $g_1, g_2, \ldots, g_n$ are called the ———— functions of $G$.

page 114



**Answer ( Please select your correct option )**

- bijective
- surjective
- component
- None of these

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**MTH631** Real Analysis II

**Question No : 4 of 26**

The quotient of two continuous functions is a ———— function wherever the denominator is non-zero.



**Answer ( Please select your correct option )**

- differentiable
- continuous
- composite
- None of these

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**MTH631** Real Analysis II

**Question No : 3 of 26**

The set $S = ((x,y), x^2 + y^2 \leq 1 \text{ or } x^2 + y^2 \geq 4)$ is not a region in $R^2$, since it is ———

page 90

**Answer ( Please select your correct option )**

- not connected
- connected
- polygonally connected
- None of these

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MTH631 Real Analysis II

Question No : 2 of 25

A set S is polygonally connected if every pair of points in S can be connected by a polygonal path lying ————

page 89

Answer ( Please select your correct option )

- outside the set S
- partially in S
- entirely in S
- None of these

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**MTH631** Real Analysis II

**Question No : 1 of 25**

A set is closed if and only if it always contains all of its ——

page 86

**Answer ( Please select your correct option )**

- limit points
- neighbourhood points
- interior points
- None of these

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**MTH631** Real Analysis II

**Question No : 20 of 26**

If $f$ and $g$ are integrable on $S$ and $f(X) \leq g(X)$ for $X$ in $S$, then ———

page163

**Answer ( Please select your correct option )**

- $\int_S f(X)dX \geq \int_S g(X)dX$
- $\int_S f(X)dX \leq \int_S g(X)dX$ ⇦
- $\int_S f(X)dX > \int_S g(X)dX$
- None of these

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**MTH631** Real Analysis II

**Question No : 19 of 25**

The lower integral $\underline{\int_R} f(X)d(X)$ of $f$ over $R$ is the supremum of all the ——

page 135

**Answer ( Please select your correct option )**

- lower sums
- upper sums
- average sums
- None of these

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**MTH631** Real Analysis II

**Question No : 19 of 25**

The upper integral $\overline{\int_a^b} f(X)d(X)$ of $f$ over $R$ is the infimum of all the———

page 135

**Answer ( Please select your correct option )**

- lower sums
- upper sums
- average sums
- None of these

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**MTH631** Real Analysis II

**Question No : 17 of 26**

A function $f$ is said to be absolutely integrable on $[a,b)$ if $f$ is locally integrable on $[a,b)$ and ———

page 145



**Answer ( Please select your correct option )**

- $\int_a^b |f(x)|dx < \infty$
- $\int_a^b |f(x)|dx = \infty$
- $\int_a^b |f(x)|dx \geq \infty$
- None of these

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**MTH631** Real Analysis II

**Question No : 15 of 26**

If $f$ and $g$ are locally integrable on $[a,b)$, $g(x) > 0$ and $f(x) \geq 0$ on some subintervals $[a_1,b)$ of $[a,b)$, and $\lim_{x \to b^-} \frac{f(x)}{g(x)} = M$. If $M = 0$ and $\int_a^b g(x)dx < \infty$, then

page 144

**Answer ( Please select your correct option )**

- $\int_a^b f(x)dx$ and $\int_a^b g(x)dx$ converges together
- $\int_a^b f(x)dx = \infty$
- $\int_a^b f(x)dx < \infty$
- $\int_a^b f(x)dx \geq \infty$

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**MTH631** Real Analysis II

**Question No : 15 of 26**

The function $f(x) = (1-x)^{-p}$ is locally integrable on ————

page 139

**Answer ( Please select your correct option )**

- [0,1]
- [-1,1)
- (0,1)
- [0,1)

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**MTH631** Real Analysis II

**Question No : 14 of 25**

If $p(X)$ is a homogeneous polynomial of degree $r$ in $X - X_0$. If $p(X) \leq 0$ for all $X \neq X_0$. Then $p$ is called _____

page 134



**Answer ( Please select your correct option )**

- positive semidefinite
- positive definite
- negative semidefinite
- negative definite

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Start Time 11:05 AM

**MTH631** Real Analysis II

**Question No : 13 of 26**

If value of a becomes zero in the Taylor series $\sum_{n=0}^{\infty} \frac{f^n(a)}{n!}(x-a)^n$, then the series is called ——

Page 44

**Answer ( Please select your correct option )**

- Maclaunn series
- Founer series
- Power series
- None of these

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**MTH631** Real Analysis II

**Question No : 13 of 26**

If value of a becomes zero in the Taylor series $\sum_{n=0}^{\infty} \frac{f^n(a)}{n!}(x-a)^n$, then the series is called ——

Page 44

**Answer ( Please select your correct option )**

- Maclaunn series
- Founer series
- Power series
- None of these

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**MTH631** Real Analysis II

**Question No : 12 of 25**

If $h = f(r, \theta)$, then using the chain rule, $\frac{\partial h}{\partial r} =$ _____

Page 127

**Answer ( Please select your correct opt**

- $\frac{\partial f}{\partial r}\cdot\frac{\partial r}{\partial x}+\frac{\partial f}{\partial r}\cdot\frac{\partial r}{\partial y}$
- $\frac{\partial f}{\partial x}+\frac{\partial x}{\partial r}+\frac{\partial f}{\partial y}+\frac{\partial y}{\partial r}$
- $\frac{\partial f}{\partial x}\cdot\frac{\partial x}{\partial r}+\frac{\partial f}{\partial y}\cdot\frac{\partial y}{\partial r}$ ⇦
- None of these

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**MTH631** Real Analysis II

**Question No : 11 of 26**

If $f(x) = 3x + 5x^3$, then at $x = -1$, $df =$ _____



**Answer ( Please select your correct option )**

- 18dy
- 18dx
- 18
- None of these

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**MTH631** Real Analysis II

**Question No :** 1 of 26

The singleton set is always —

page 81

**Answer** ( Please select your correct option )

- disconnected
- polygonally disconnected
- connected
- None of these

MTH631 Real Analysis II

Question No : 2 of 26

Any polygonally connected set either it is open or not, is always _____

page 82

Answer ( Please select your correct option )

- connected
- disconnected
- a region
- None of these

**MTH631** Real Analysis II

**Question No :** 3 of 26 **Marks:** 1 (Budgeted Ti

For the function $f(x,y)=\frac{xy}{x^2+y^2}$, the limit of $f(x,y)$ as $(x,y)\to(0,0)$ along the line $y=x$ is ——

page 87

**Answer** ( Please select your correct option )

- $\frac{1}{2}$
- $-\frac{1}{2}$
- 0
- 1

**MTH631** Real Analysis II

**Question No : 1 of 26** **Marks: 1 (Budgeted**

If $f$ and $g$ are continuous functions at the point $X_0$ in $R^*$, then $\frac{f}{g}$ is continuous at $X_0$ provided that _____

page 100

**Answer** ( Please select your correct option )

- $g(X_0) = 0$
- $g(X_0) \neq 0$
- $f(X_0) = g(X_0) = 0$
- None of these

MTH631 Real Analysis II

Question No : 5 of 26 Marks: 1 (Budgeted Ti

Suppose $U_0 \in T$ is a limit point of $T$. $g$ is continuous at $U_0$ and $f$ is continuous at $X_0 = G(U_0)$. Then ——— is continuous at $U_0$.

page 102

Answer ( Please select your correct option )

- $f \circ g$
- $\frac{f}{g}$
- $\frac{g}{f}$
- None of these

MTH631 Real Analysis II

Question No : 6 of 26

Marks: 1 (Budgeted T

A set A ⊆ R of real numbers is ———— if there exists a real number M ⊆ R, such that $x \leq M$ for every $x \in A$.

Answer ( Please select your correct option )

- bounded below
- uniformly continuous
- bounded above
- None of these

**MTH631** Real Analysis II

**Question No :** 7 of 26 **Marks:** 1 (Bud

A function $f$ is ———— on a subset $S$ of $R^n$ if $S$ is contained in an open set on which $f_1, f_2, \ldots f_k$ are continuous

page 116

**Answer** ( Please select your correct option )

- continuously differentiable
- uniformly convergent
- piecewise convergent
- None of these

MTH631 Real Analysis II

Question No : 8 of 26 Marks: 1 (Budgeted

A critical point at which a function attains its minimum value among all points where it is defined is called _____

Answer ( Please select your correct option )

- global maximum
- global minimum
- supremum
- None of these

MTH631 Real Analysis II

Question No : 9 of 26 Marks: 1 (Budgete

The rate of change of $f(x,y)$ in the direction of a unit vector is called the ———— of $f(x,y)$

Answer ( Please select your correct option )

- differential
- directional derivative
- mixed derivative
- None of these

MTH631 Real Analysis II

Question No : 18 of 26

A function of the form $L(X) = m_1 x_1 + m_2 x_2 + \ldots + m_n x_n$ for some constants $m_i$'s is called ———

page 112

Answer ( Please select your correct option )

- linear function
- quadratic function
- constant function
- None of these